

An implementation note on HCAPO:

how a wrong ρ denominator manufactured a length runaway —
and why the paper-correct form ignites one too

Agentic-RL-CA project

2026-07-29, updated 2026-07-30 (step-200 falsification)

Abstract

While ablating HCAPO (arXiv:2603.08754) on SearchQA (Qwen3-4B, non-thinking, 2048-token response cap), our runs reproduced a severe response-length runaway — mean length inflating $15\times$ (89→1642 tokens/turn) with 78% of turns clipping at the cap — while accuracy stayed flat. Per-turn instrumentation of the hindsight ratio ρ traced the cause to a single line: **our port divided by the on-policy probability of the action, where the paper divides by the intra-trajectory mean of the hindsight probabilities** (Eq. 7). The wrong ratio is systematically < 1 and, because only its numerator saturates with response length while the $1/n_{tok}$ normalisation keeps growing, it *rewards verbosity*: emitting more tokens mechanically raises ρ , hence the advantage. We derive the mechanism, verify it in a controlled synthetic test (legacy ρ moves $+0.13$ when responses lengthen $16 \rightarrow 64$ tokens; the paper form moves 0.0), and show the corrected implementation restores every property the paper assumes: ρ centred on 1.0, two-sided, and discriminating between turns for the first ~ 150 steps. **Update (step 200): the corrected run then ignited the same runaway** (221 $\rightarrow$ 1232 tokens between steps 150 and 200, clip 0.51) — falsifying our structural prediction. The ρ instrumentation identifies the shared root: Eq. 6’s *per-token-mean* scoring is pulled toward generic fluency as turns lengthen, so a below-mean turn can raise its ρ by lengthening; the intra-trajectory normalisation cancels uniform inflation but not this transient, and the across-turn spread collapses (σ : 0.135 $\rightarrow$ 0.030) until the hindsight term is inert. On this task the published formulation, not merely our port of it, carries a length instability. All prior HCAPO numbers from this codebase characterise the buggy variant; the corrected run characterises the published method on SearchQA.

1 The two formulas

What the paper specifies (arXiv:2603.08754, Eq. 6–7). Per turn t of a trajectory with T turns, with s_{final} the trajectory’s last observation injected into the prompt:

$$\pi_{hind}(a_t) = \exp\left(\frac{1}{T_{temp}|a_t|} \sum_j \log \pi_\theta(y_j \mid y_{<j}, s_t, s_{final})\right) \quad (\text{Eq. 6})$$

$$\rho_t = \text{clip}\left(\pi_{hind}(a_t) / \bar{\pi}_{hind}, C_{min}, C_{max}\right), \quad \bar{\pi}_{hind} = \frac{1}{T} \sum_{k=1}^T \pi_{hind}(a_k) \quad (\text{Eq. 7})$$

The denominator is the **intra-trajectory mean over that trajectory’s turns** — the paper calls it “group-normalization across actions within the same episode”. By construction ρ is centred on 1.0: pivotal turns score > 1 , redundant turns < 1 , and the symmetric clip $[C_{min}, C_{max}] = [0.8, 1.2]$ is well-specified around that centre. ρ then scales a discounted return, $Q_t^H = \rho_t \gamma^{T-1-t} R$, which is group-normalised across the G rollouts of the prompt and added (weight ω) to the standard GRPO term.

What we implemented (gigpo/core_hcapo.py, pre-2026-07-29):

```

mean_logratio = ((hindsight_log_probs - policy_log_probs) * mask).sum(-1) / ntok
rho = exp(mean_logratio / t_temp).clamp(0.8, 1.2)
# == pi_hind(a_t) / pi_POLICY(a_t)      <- divided by the ON-POLICY prob of the
      action

```

The numerator matches Eq. 6 exactly (per-token-normalised, temperature-sharpened hindsight log-prob). The denominator does not: instead of comparing each turn’s hindsight score against its *siblings in the same trajectory*, it compares hindsight against on-policy conditioning of the *same* tokens. That ratio measures something else entirely: *how much the injected s_{final} text perturbs the model off-distribution* — which is why, measured over 325 training steps, our ρ was **never once above** 1.0 (max 0.9986; the upper clip literally never bound) while 10.5% of turns pinned at the 0.8 floor.

2 Why the wrong ratio rewards verbosity

Write the per-token log-ratio $\delta_j = \log \pi(y_j | y_{<j}, s, s_{final}) - \log \pi(y_j | y_{<j}, s)$, so `mean_logratio` = $\frac{1}{n_{tok}} \sum_j \delta_j$. If δ_j had a position-independent mean, n_{tok} would cancel and length would not matter. It does not: the hindsight text perturbs the *prompt*, and both conditionings share the identical generated prefix $y_{<j}$, so δ_j is large (negative) on the first tokens — where the prompt dominates the context — and decays toward 0 as the response’s own text takes over. The numerator therefore **saturates** at some constant C^- while the divisor grows:

$$\text{mean_logratio} \approx \frac{C^-}{n_{tok}} \xrightarrow{n_{tok} \uparrow} 0^-, \quad \Rightarrow \quad \rho = \exp(\text{mean_logratio}/T_{temp}) \rightarrow 1^-.$$

Longer responses $\Rightarrow$ higher $\rho \Rightarrow$ higher Q^H .

Two independent confirmations:

(a) The training data itself. Inverting the logged ρ to the implied *total* $\sum_j \delta_j = T_{temp} \ln(\rho) n_{tok}$: at step 25 ($n_{tok}=89, \rho=0.909$) the sum is -42.5 ; at step 300 ($n_{tok}=1635, \rho=0.996$) it is -36.0 . **Length grew 18 $\times$; the total hindsight effect did not grow at all** (Fig. 1, left). The entire rise of ρ is the denominator. (Order-of-magnitude arithmetic: early values are censored by the 0.8 clip floor, and inverting a batch mean through exp is Jensen-approximate.)

(b) A controlled synthetic test. Fix a front-loaded perturbation ($\delta_j = -2$ on the first 8 tokens, 0 after) and vary only response length (Fig. 2): the legacy ρ climbs from 0.82 toward 1.0 as n_{tok} grows (+0.13 between 16 and 64 tokens); the paper’s ρ is **exactly length-invariant** — numerator and denominator share the same per-token-normalised functional form, so any uniform length change cancels. Note this immunity does *not* depend on the front-loading diagnosis being right, which is a good property.

How a ρ bias becomes a length ratchet. Q^H feeds a group normalisation across the $G=5$ rollouts of each prompt, which is invariant to uniform scaling — so “everyone gets longer, everyone gains” cancels. The gradient pressure is *relative*: among sibling rollouts of the same prompt, the **longer rollout has higher ρ than its siblings**, out-scores them, and is reinforced. Each iteration the whole group lengthens, but the relative pressure persists — a ratchet. This predicts both observed dynamics (Fig. 3): a *phase transition* rather than smooth drift (legacy A: 551 $\rightarrow$ 1371 tokens in 25 steps once the loop won), and *self-termination* as ρ approaches its ceiling and the across-sibling spread compresses (legacy A: ρ std 0.055 $\rightarrow$ 0.038; length plateaus at ~ 1635 against the 2048 cap, clip 0.77–0.78, *above* the 0.754 that accompanied the 1.7B thinking-mode collapse). Halving the hindsight weight ($\omega=0.5$, legacy B) shrinks the step size along the same

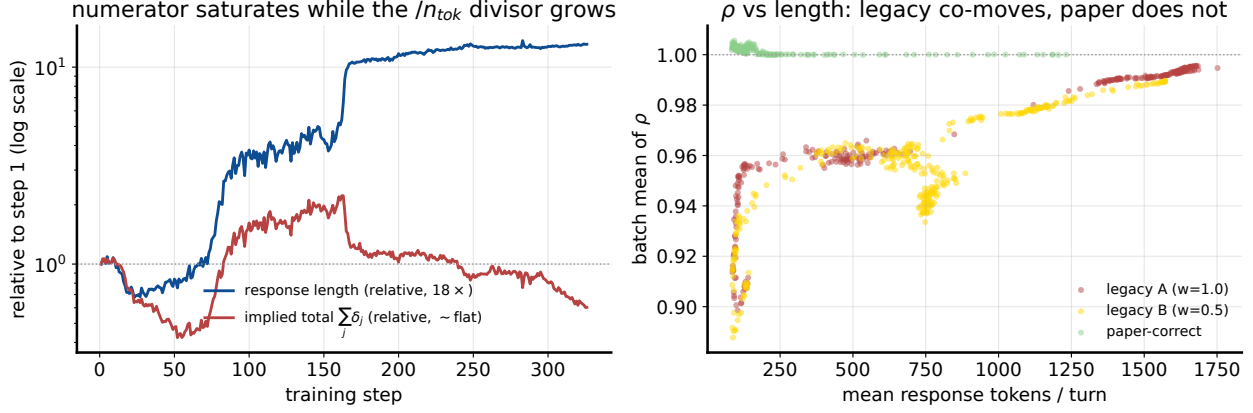


Figure 1: **The dilution mechanism, measured.** *Left:* in the legacy $\omega=1$ run, the implied total hindsight log-ratio $\sum_j \delta_j$ (red) is $\sim$ flat while response length (blue) grows $18\times$ — ρ rises purely because the $1/n_{tok}$ divisor grows. *Right:* batch-mean ρ against response length: the two legacy runs trace a tight monotone curve toward 1.0; the paper-correct run sits at $\rho \approx 1$ independent of length.

gradient — it **delays the runaway by ~ 125 steps but does not prevent it**. Throughout, accuracy stayed flat (≈ 0.39 macro-EM): the verbosity was *waste*, not fuel.

3 The corrected implementation

`gigpo/core_hcapo.py` now implements Eq. 6–7 as the default (`rho_denominator="intra_traj_mean"`; the buggy path is retained under `"on_policy"` solely to reproduce the withdrawn runs):

```
# Eq. 6: per-token-normalised, temperature-sharpened hindsight log-prob
log_pi_hind = (hindsight_log_probs * mask).sum(-1) / ntok / t_temp
# Eq. 7: self-normalise by the mean over the TURNS of the same trajectory
for tu in unique(traj_uid):
    idx = rows_of(tu); rep = one_row_per_distinct_turn(idx)
    xm = log_pi_hind[rep].max() # numerical stability shift
    rho[idx] = exp(log_pi_hind[idx] - xm) / mean(exp(log_pi_hind[rep] - xm))
rho = clip(rho, 0.8, 1.2)
```

The policy log-probs no longer enter ρ at all, matching the paper. Unit tests: (i) the synthetic length test above (legacy $+0.1325$, paper 0.0000); (ii) ρ centred on 1.0 with two-sided spread on toy trajectories.

4 The corrected form: 150 clean steps, then the same runaway

The structural argument we made — and where it fails. We argued the ratchet needs a way for a rollout to gain by being globally more verbose, and that Eq. 7 removes it: numerator and denominator share the $1/|a|$ normalisation, so *uniform* verbosity cancels identically. That argument is correct as far as it goes — and was **falsified at step 200** because the gradient it ignores is the *transient, non-uniform* one. Eq. 6 scores a turn by its *per-token mean* log-prob, and a long turn’s mean is dominated by generically-predictable continuation tokens. A turn currently *below* its trajectory’s mean can therefore raise its ρ simply by lengthening — diluting its distinctive

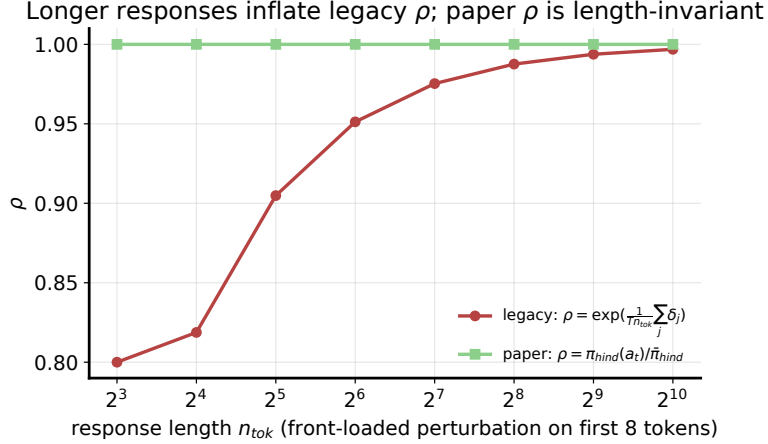


Figure 2: **Controlled test of the mechanism.** Front-loaded perturbation ($\delta = -2$ on the first 8 tokens), response length varied, nothing else. Legacy ρ (red) inflates with length; the paper’s self-normalised ρ (green) is exactly 1 at every length.

(low-probability) early tokens toward generic fluency. Lengthening is never uniform while this is happening, so the intra-trajectory normalisation does not cancel it; the process self-terminates only when all turns have homogenised ($\rho \approx 1$ everywhere) and the hindsight signal is dead.

Empirical. The first ~ 150 steps looked exactly as the paper intends (single seed; per-step instrumentation):

	legacy (withdrawn), step 1	paper-correct, step 1
ρ mean	0.9078	1.0006
ρ min / max	0.800 / 0.9986	0.800 / 1.2000
frac at lower clip	0.105	0.023
frac at upper clip	0.000	0.014
$ \text{macro} : \text{micro} $ (adv. terms)	1 : 1.40	1 : 1.14

Every property the paper assumes is restored: ρ is centred on 1.0 and **two-sided** — **the upper clip binds for the first time in this codebase** (1.4% of turns), i.e. some turns are now scored as *pivotal*, which is the entire point of hindsight credit and was structurally impossible under the legacy ratio. The spread is *growing* (std 0.073 $\rightarrow$ 0.143 over 45 steps) where the legacy spread collapsed (0.055 $\rightarrow$ 0.038): the signal differentiates turns instead of saturating (Fig. 4). Behaviourally, at step 25: macro-EM 0.352, clip 0.001, response length 91 tokens — indistinguishable from GRPO at the same point, as expected.

Then the ignition. Length: 147@100 $\rightarrow$ 221@150 $\rightarrow$ 249@160 $\rightarrow$ 412@170 $\rightarrow$ 611@180 $\rightarrow$ 985@190 $\rightarrow$ **1232@200** (clip 0.505); EM flat across it (0.401 $\rightarrow$ 0.417 $\rightarrow$ 0.408) — the same phase-transition-then-waste signature as the legacy runs, ~ 40 steps later than legacy A. Concurrently the ρ spread *collapsed* (σ : 0.135@100 $\rightarrow$ 0.068@150 $\rightarrow$ 0.030@200, both clip fractions $\rightarrow$ 0): by step 200 every turn scores $\rho \approx 1$ and the hindsight term is inert — HCAPO has degenerated to GRPO plus a positional discount γ^{T-1-t} , at $8\times$ GRPO’s token cost. *Interpretation caveats:* single seed; SearchQA trajectories have $T \leq 4$ turns (a noisy 4-sample mean in Eq. 7) and our s_{final} is a long retrieved-documents block — the paper’s own tasks (ALFWorld/WebShop: short action-style turns, 512-token caps, Qwen2.5-7B) may never enter this regime, so this is a statement about the formulation *on this task class*, not a refutation of the paper’s results. The decisive within-batch probe — per-sample $\text{corr}(\rho, |a_t|)$ before vs during ignition — is now the top instrumentation priority.

Legacy HCAPO runs away in length; the paper-correct run (so far) does not

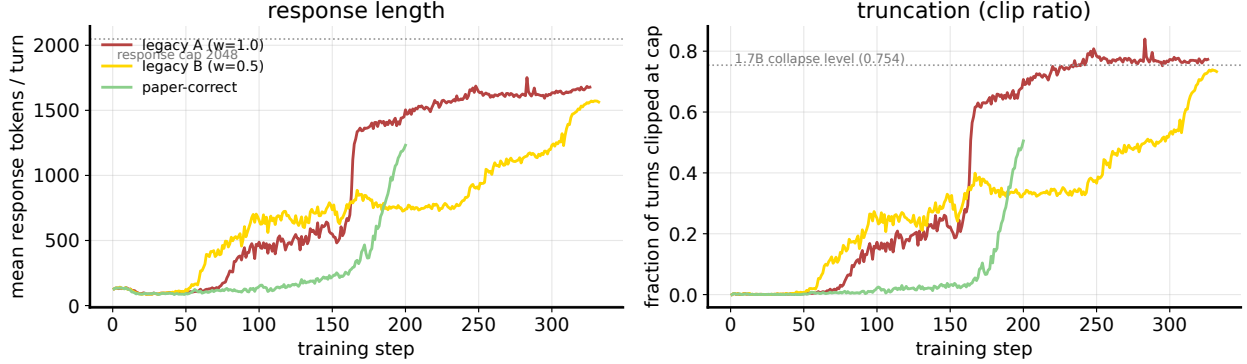


Figure 3: **The runaway, and its absence so far under the fix.** Mean response length (left) and cap-clip fraction (right) per training step. Legacy $\omega=1$ (red) undergoes a phase transition at $\sim$ step 160 and saturates against the 2048 cap; $\omega=0.5$ (orange) follows the same path $\sim$ 125 steps later; the paper-correct run (green) holds normal length for $\sim$ 150 steps — then ignites the same phase transition $\sim$ 40 steps after legacy A, reaching 1232 tokens by step 200.

The run continues to 500 (abort rule requires $EM < 0.30$; it is 0.408); final update at 500.

5 Record

Withdrawn: both legacy 4B runs (killed at steps $\sim$ 324/330), and all earlier HCAPO numbers from this codebase — SearchQA 1.7B (0.357 @ clip 0.754), 4B legacy arms ($\approx$ 0.39), AlfWorld 1.7B (0.791) — as characterisations of the *published* method (they remain valid measurements of the buggy variant). The full timestamped trail (pre-registration, amendments, retraction of two premature trend calls, and this correction) is in `docs/reports/2026-07-21_4b_horizon_eval/results/hcapo_wave3_preregistration.md`. Provenance caveat: Eq. 6–7 were transcribed from the arXiv HTML via automated fetch (re-queried twice, quoted verbatim); confirm against the PDF before external use. Rebuild: `assets/build_figures.py` (parses the three training logs directly), then `tectonic report.tex`.

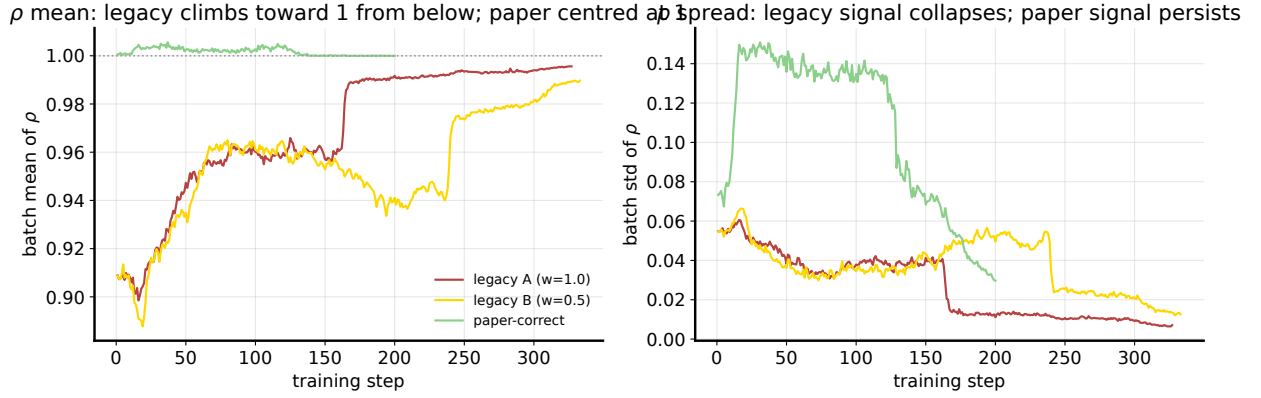


Figure 4: ρ statistics per training step. *Left:* legacy ρ (red/orange) starts at 0.91 and is crushed toward 1.0 from below as length inflates; the paper-correct run (green) sits at 1.0 from step one. *Right:* legacy spread collapses as ρ saturates toward 1 from below; the corrected spread grows for ~ 100 steps (discriminating between turns) and then *also* collapses ($0.135 \rightarrow 0.030$) as lengthening homogenises the per-token means — the spread, not the mean, is where the corrected form’s runaway is visible.